Resultant amplitude is maximum when,

$\phi = 0, \cos 0 = 1, A_{max} = \sqrt{a_1^2 + a_2^2 + 2a_1a_2}$

$A_{max} = \sqrt{(a_1 + a_2)^2} = \sqrt{(5+3)^2} = \sqrt{(8)^2}$
$= 8 \ units$

Resultant amplitude is minimum when,

$\phi = \pi, \cos \pi = -1, A_{max} = \sqrt{a_1^2 + a_2^2 - 2a_1a_2}$

$A_{min} = \sqrt{(a_1 - a_2)^2} = \sqrt{(5-3)^2} = \sqrt{(2)^2}$
$= 2 \ units$

$I \propto A^2$

$\frac{I_{max}}{I_{min}} = \frac{(A_{max})^2}{(A_{min})^2}$

Substituting,

$\frac{I_{max}}{I_{min}} = \frac{(8)^2}{(2)^2} = \frac{64}{4} = 16$ (or)

$I_{max} : I_{min} = 16 : 1$

EXAMPLE 7.3

Two light sources of equal amplitudes interfere with each other. Calculate the ratio of maximum and minimum intensities.

Solution

Let the amplitude be a.

The intensity is, $I \propto 4a^2 \cos^2(\phi/2)$

or $I = 4I_0 \cos^2(\phi/2)$

Resultant intensity is maximum when,

$\phi = 0, \cos 0 = 1, I_{max} \propto 4a^2$

Resultant amplitude is minimum when,

$\phi = \pi, \cos(\pi/2) = 0, I_{min} = 0$

$I_{max} : I_{min} = 4a^2 : 0$

EXAMPLE 7.4

Two light sources have intensity of light as I_0. What is the resultant intensity at a point where the two light waves have a phase difference of $\pi/3$?

Solution

Let the intensities be I_0.

The resultant intensity is, $I = 4 I_0 \cos^2(\phi/2)$

Resultant intensity when, $\phi = \pi/3$, is $I = 4I_0 \cos^2(\pi/6)$

$I = 4I_0 (\sqrt{3}/2)^2 = 3I_0$

7.3.1 Phase difference and path difference

Phase is the angular position of vibration when a wave is progresses, there is a relation between the phase of the vibration and the path travelled by the wave. We can express the phase in terms of path and vice versa. In the path of the wave, one wavelength λ corresponds to a phase of 2π as shown in Figure 7.7. A path difference δ corresponds to a phase difference ϕ as given by the equation,

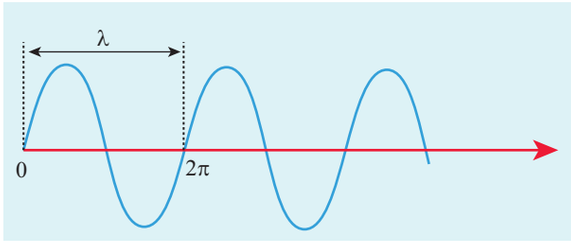


Figure 7.7 Path difference and phase difference

$$\delta = \frac{\lambda}{2\pi} \times \phi \text{ (or) } \phi = \frac{2\pi}{\lambda} \times \delta \qquad (7.23)$$

For constructive interference, the phase difference should be, $\phi = 0, 2\pi, 4\pi \ldots$ Hence,

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